

Question Paper: Machine Learning

****Machine Learning Question Paper****

****Instructions:****

- * The question paper consists of three sections: Multiple Choice Questions, Short Questions, and Long/Descriptive Questions.
- * Attempt all questions in the given time frame.
- * Use a pencil to fill in the MCQs and a pen to write the short and long answers.

****Section A: Multiple Choice Questions (1-10)****

1. What is the primary goal of machine learning?

- A) To develop software that can think like humans
- B) To develop algorithms that can learn from data
- C) To develop hardware that can process large amounts of data
- D) To develop networks that can communicate with humans

Correct answer: B) To develop algorithms that can learn from data

2. Which of the following machine learning algorithms is used for classification problems?

- A) Linear Regression
- B) Decision Trees
- C) Clustering
- D) All of the above

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Correct answer: B) Decision Trees

3. What is the term for the process of dividing a dataset into training and testing sets?

- A) Cross-validation
- B) Overfitting
- C) Underfitting
- D) Data preprocessing

Correct answer: A) Cross-validation

4. Which of the following is a type of supervised learning?

- A) Unsupervised learning
- B) Reinforcement learning
- C) Classification
- D) Clustering

Correct answer: C) Classification

5. What is the purpose of the activation function in a neural network?

- A) To introduce non-linearity into the model
- B) To reduce the dimensionality of the data
- C) To increase the complexity of the model
- D) To decrease the accuracy of the model

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Correct answer: A) To introduce non-linearity into the model

6. Which of the following is a popular deep learning framework?

- A) TensorFlow
- B) PyTorch
- C) Keras
- D) All of the above

Correct answer: D) All of the above

7. What is the term for the process of adjusting the model's parameters to minimize the loss function?

- A) Backpropagation
- B) Forward propagation
- C) Optimization
- D) Regularization

Correct answer: A) Backpropagation

8. Which of the following is a type of machine learning model that can learn from sequential data?

- A) Recurrent Neural Network (RNN)
- B) Convolutional Neural Network (CNN)
- C) Autoencoder

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D) Generative Adversarial Network (GAN)

Correct answer: A) Recurrent Neural Network (RNN)

9. What is the purpose of regularization in machine learning?

A) To increase the complexity of the model

B) To reduce the overfitting of the model

C) To increase the accuracy of the model

D) To decrease the computational cost of the model

Correct answer: B) To reduce the overfitting of the model

10. Which of the following is a popular metric for evaluating the performance of a classification model?

A) Mean Squared Error (MSE)

B) Mean Absolute Error (MAE)

C) Accuracy

D) F1-score

Correct answer: C) Accuracy

****Section B: Short Questions (11-15)****

11. Explain the concept of overfitting in machine learning. (3 marks)

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12. Describe the difference between supervised and unsupervised learning. (4 marks)
13. What is the purpose of data preprocessing in machine learning? Provide an example. (4 marks)
14. Explain the concept of bias-variance tradeoff in machine learning. (4 marks)
15. Describe the architecture of a simple neural network. (5 marks)

****Section C: Long/Descriptive Questions (16-18)****

16. Explain the concept of deep learning and its applications in computer vision. Provide an example of a deep learning model used for image classification. (8 marks)
17. Describe the process of building a machine learning model using the CRISP-DM methodology. Explain each step in detail. (10 marks)
18. Explain the concept of reinforcement learning and its applications in robotics. Provide an example of a reinforcement learning algorithm used for robotic control. (10 marks)

****Answer Key****

****Section A: Multiple Choice Questions****

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1. B) To develop algorithms that can learn from data
2. B) Decision Trees
3. A) Cross-validation
4. C) Classification
5. A) To introduce non-linearity into the model
6. D) All of the above
7. A) Backpropagation
8. A) Recurrent Neural Network (RNN)
9. B) To reduce the overfitting of the model
10. C) Accuracy

****Section B: Short Questions****

11. Overfitting occurs when a model is too complex and learns the noise in the training data, resulting in poor performance on unseen data.
12. Supervised learning involves learning from labeled data, while unsupervised learning involves learning from unlabeled data.
13. Data preprocessing involves cleaning, transforming, and feature engineering the data to prepare it for modeling. Example: Handling missing values by imputing them with the mean or median.
14. The bias-variance tradeoff refers to the tradeoff between the model's ability to fit the training data (bias) and its ability to generalize to unseen data (variance).

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15. A simple neural network consists of an input layer, one or more hidden layers, and an output layer, with each layer consisting of multiple neurons that apply an activation function to the input.

****Section C: Long/Descriptive Questions****

16. Deep learning refers to the use of multiple layers in a neural network to learn complex patterns in data. Example: Convolutional Neural Networks (CNNs) are used for image classification tasks such as object detection and image recognition.

17. The CRISP-DM methodology involves the following steps: business understanding, data understanding, data preparation, modeling, evaluation, and deployment. Each step involves a series of tasks that help build a machine learning model.

18. Reinforcement learning involves learning from interactions with an environment to achieve a goal. Example: Q-learning is a reinforcement learning algorithm used for robotic control tasks such as navigation and manipulation.